

开场介绍 (Opening)

Hello, everyone.

We are **Team 3**, and today we are going to present our project titled

“Application of Fuzzy Logic (Tsukamoto System) in Automotive Braking.”

Our team members are as follows

In this presentation, we'll briefly review the idea of fuzzy logic,
then focus on the **Tsukamoto System**,
and finally explain how it can be applied in
an **automotive braking control** case.

第一部分：引入 (Introduction)

Since we already have a good understanding of how **fuzzy logic** works, we'll now move directly to the **Tsukamoto System** —

We'll also look at how this system can be used in a **car braking example**,
where the car adjusts brake force based on **speed** and **distance**.

第二部分：Tsukamoto System 讲解

The **Tsukamoto System** is a special type of fuzzy inference system.

Its main feature is the ability to **directly obtain crisp outputs** through a simple weighted average, eliminating the need for additional “defuzzification” steps.

This is achieved by using **monotonic membership functions** for the consequents (抗sei困死) —

meaning that each output function is either increasing or decreasing.

Thanks to this design, the system can use **algebraic inversion**
to easily compute the exact output for each rule.

This structure makes Tsukamoto very **efficient and interpretable**,
and that's why it is widely used in control systems —
including **automotive braking example**.

Here are its **five main steps**:

The following will provide a detailed introduction.

1 Fuzzification – Convert input values into membership degrees. Triangular membership functions are employed.

2 Rule Formation – Define linguistic rules

if A and B then C

like “IF speed is high AND distance is near THEN brake is strong.”

3 Inference – Compute the firing strength α of each rule.

α is obtained by taking the minimum.

4 Output Mapping – Invert the function to get crisp z_i values.

5 Defuzzification – Use weighted average:

$$z^* = \sum(\alpha_i \cdot z_i) / \sum \alpha_i.$$

Now let's apply this system to a real case — automotive braking.

Tsukamoto 系统是一种特殊的模糊推理模型。

它的最大特点是通过加权平均可以**直接得到精确输出 (crisp output)**，而不需要额外的“去模糊化”步骤。

这是因为它的输出隶属函数被设计成**单调函数**，

即要么递增，要么递减。

这样我们就能通过**代数反演**的方法，

直接求出每条规则对应的输出值。

这种结构让 Tsukamoto 模型既**高效又可解释**，

因此被广泛用于控制系统中，

包括**汽车制动案例**。

它的运行流程主要分为五个步骤：

- 1 模糊化**：把输入数值转化为隶属度；
- 2 规则形成**：建立语言规则，例如“如果速度高且距离近，则刹车强”；
- 3 推理计算**：得到每条规则的激活强度 α ；
- 4 输出映射**：通过单调函数反演求出每条规则的输出值 z_i ；
- 5 去模糊化**：加权平均得到最终输出： $z^* = \sum(\alpha_i \cdot z_i) / \sum \alpha_i$ 。